

Development of a metabolic modelling workflow for the study of microbial communities

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Project repository: <https://github.com/brunosate21-a11y/Projeto-.git>

Abstract. This work describes the development and application of a workflow implemented in Snakemake for the automated reconstruction of genome-scale metabolic models (GEMs) from previously assembled genomes, with a focus on metagenome-assembled genomes (MAGs), and for the subsequent simulation of metabolic interactions at the community level. The pipeline integrates, within a single and traceable flow, the stages of genomic quality control (with CheckM or, in a bypass mode, CheckM2), automatic GEM reconstruction with CarveMe, model validation with Memote, and simulation of community interactions with three complementary tools (SMETANA, SteadyCom, and MICOM). The workflow was validated on a set of four reference genomes (*Bacillus subtilis* 168, *Escherichia coli* K-12, *Pseudomonas putida* NBRC 14164, and *Streptococcus pneumoniae* TIGR4) and subsequently applied to a real set of 22 prokaryotic MAGs from an environmental sample, annotated with Bakta and assessed with CheckM2. The pipeline produced functional models for all genomes and recovered a dense interaction structure in the real community: a high metabolic-niche overlap, quantified by the SMETANA Metabolic Resource Overlap index ($MRO \approx 0.72$), and more than two hundred potential cross-feeding pairs. A per-organism analysis further revealed that relative abundance and metabolic activity are weakly, even negatively, correlated, so that the most abundant organisms are not necessarily the metabolically most active. The code is versioned and documented, promoting reproducibility and reuse.

Keywords: Genome-Scale Metabolic Models (GEMs) · Flux Balance Analysis (FBA) · Metabolic Interactions · Computational Pipeline · Snakemake · MAGs

1 Background and motivation

Shotgun metagenomics has transformed the study of microbial communities by enabling access to the genetic content of microorganisms directly from environmental or clinical samples, without cultivation [1,2]. From these data it is possible to reconstruct metagenome-assembled genomes (MAGs), draft genomes for the (often uncultured) organisms present in a community and a basis for downstream functional analyses [2,3]. Among these, genome-scale metabolic modelling is

particularly relevant, as it builds models that describe the metabolic potential of each organism and support *in silico* simulations. Considered jointly, these models make it possible to study emergent community properties such as competition for nutrients, metabolite sharing, and other ecological interactions.

Several tools and pipelines already address parts of this problem. metaGEM, for example, reconstructs genome-scale metabolic models directly from metagenomes and links them to community simulation [4]. In practice, however, the individual stages (quality control, reconstruction, model validation, and community simulation) are frequently carried out with separate programs that have heterogeneous formats, dependencies, and methodological assumptions. Integrating these stages into a single, traceable, and reproducible process is the gap that this work addresses. Rather than re-implementing every stage of metagenomic analysis, the proposal focuses on the phases that follow assembly, from previously obtained and assessed prokaryotic MAGs (a constraint imposed by the metabolic reconstruction tool used, CarveMe [11]) through to GEM reconstruction and community-level simulation.

1.1 Objectives

The central objective of this work was to develop and apply a modular and reproducible workflow, implemented in Snakemake [5,6], for the metabolic modelling of microbial communities from previously assembled genomes, with a focus on MAGs. Covering the project as a whole, the specific objectives were: (i) to design and implement, in Snakemake, an integrated pipeline chaining genomic quality control, automatic GEM reconstruction, model validation, and community simulation; (ii) to validate the pipeline end to end on reference genomes with well-characterised metabolism; (iii) to apply it to a real set of environmental MAGs; (iv) to compare, on the same data, three community-simulation tools with complementary assumptions (SMETANA, SteadyCom, and MICOM), inferring patterns of competition, metabolite sharing, and community growth; and (v) to incorporate external metadata (upstream quality reports and relative abundances) into the flow, as required by a real application scenario.

2 State of the Art

The functional understanding of microbial communities relies on the integration of genomic data with metabolic modelling approaches. This section reviews the relevant domains: the reconstruction and quality assessment of MAGs from metagenomic data, the construction of genome-scale metabolic models, and metabolic modelling at the community level, with particular attention to the strengths and limitations of the available tools, which motivate the choices made in this work.

2.1 Metagenomics and MAG reconstruction

Shotgun metagenomics enabled the cultivation-independent recovery of genomes directly from environmental samples, substantially broadening the microbial diversity accessible to genomic analysis [1,2,7]. MAGs result from stages such as assembly, binning, and refinement, and their quality is a determining factor for any downstream functional or metabolic analysis. Bowers et al. proposed the MIMAG/MISAG standards, according to which completeness and contamination are the essential metrics for assessing these genomes [3]. Under these criteria, a high-quality MAG should present more than 90% completeness and less than 5% contamination, while a medium-quality MAG should present at least 50% completeness and less than 10% contamination [3]. These thresholds matter for metabolic modelling: a model can only contain reactions whose genes were recovered, so completeness below the medium-quality floor risks omitting essential reactions of central pathways, whereas contamination introduces reactions from foreign lineages and therefore spurious functional associations [8]. In this work, the minimum acceptable quality for a genome to proceed to reconstruction was set in line with these MIMAG recommendations and made configurable (see Section 3.3).

The quality of MAGs is assessed with dedicated tools. CheckM, for example, estimates completeness and contamination from sets of lineage-specific single-copy marker genes [8]. More recently, CheckM2 replaced the marker-based strategy with machine-learning models trained on synthetic and experimental genomes, offering more accurate and faster estimates, in particular for novel lineages or for reduced-size genomes where marker-based methods are less reliable [22].

2.2 Metabolic modelling and GEM reconstruction

A genome-scale metabolic model (GEM) is a structured mathematical representation of an organism’s metabolism, integrating metabolites, reactions, gene-protein-reaction associations, and biophysical constraints. These models make it possible to study metabolic capabilities, predict pathway usage, and simulate growth phenotypes [9,10]. The most widely used theoretical basis is Flux Balance Analysis (FBA), a constraint-based approach that describes metabolic fluxes by assuming an internal steady state and an objective function, frequently biomass production [9]. A rigorous GEM reconstruction is an iterative process of curation and validation [10], demanding in time and manual effort, which motivated automated tools. Among these, CarveMe enables the rapid generation of models for microbial species and communities through a top-down approach based on a curated universal model [11]. Speed and reproducibility are recognised benefits of automated reconstruction, although the resulting draft models do not replace expert manual curation. Interoperability is also central: standardised formats and consistent identifiers for reactions and metabolites are fundamental, and the BiGG Models database is particularly relevant here as a source of standardised identifiers [12]. In turn, the quality of the generated GEMs can be assessed in a standardised manner with Memote, a community-driven test suite that quantifies stoichiometric consistency and annotation coverage [23].

2.3 Community metabolic modelling

Models of individual organisms describe specific metabolic capabilities, but do not directly represent emergent properties of communities, such as competition for resources, cross-feeding, metabolic dependencies, and stable coexistence. Community-level modelling has thus become a relevant approach for investigating functional interactions between microorganisms, and several tools, with different assumptions, are available. Constraint-based methods such as OptCom, cFBA, SteadyCom, MICOM, and the Microbiome Modelling Toolbox extend FBA to multi-species systems by combining the GEMs of the members into a compartmentalised community model [17,18]. A complementary line of work, represented by SMETANA, focuses on inferring the potential for cross-feeding and metabolic interdependence rather than on predicting growth [14]. Pipelines such as metaGEM connect this modelling step to metagenomic data by reconstructing community models directly from metagenomes [4].

The choice between these tools is not arbitrary. A structured evaluation by Scott et al. assessed 24 constraint-based tools for microbial consortia, both qualitatively (against FAIR software-quality criteria) and quantitatively against experimental case studies, and found that the more up-to-date, accessible, and well-documented tools, including SteadyCom and MICOM, generally performed best in software quality and predictive capacity [13]. Guided by these findings and by the complementary nature of their assumptions, three tools were selected. SMETANA computes global indices of resource overlap (MRO) and interaction potential (MIP), as well as species-pair dependencies, and is therefore suited to characterising the *potential* for competition and cross-feeding [14,15,16]. SteadyCom predicts flux distributions and relative abundances compatible with a community-level steady state, by imposing a common growth rate on all members [17]. MICOM predicts taxon-specific growth rates and exchange fluxes from relative abundances and environmental constraints, using a cooperative trade-off between community and individual growth [18]. Together they offer complementary perspectives on interaction potential, growth, and exchange fluxes over the same community.

2.4 Workflow management and reproducibility

The construction of reproducible bioinformatics pipelines is a central requirement in analyses that chain multiple tools, parameters, and software versions [19,20]. Several workflow management systems address this need; the present work uses Snakemake, which automates analyses in a readable, scalable, and reproducible manner and is widely adopted in bioinformatics [5,6]. Its Python-based language infers dependencies and parallelisation automatically from the input and output files declared by each rule, representing the pipeline as a directed acyclic graph (DAG) [5]. Crucially for reproducibility, Snakemake also makes it possible to associate isolated Conda environments with individual stages, promoting computational isolation and fine-grained control over the software stack [6,21].

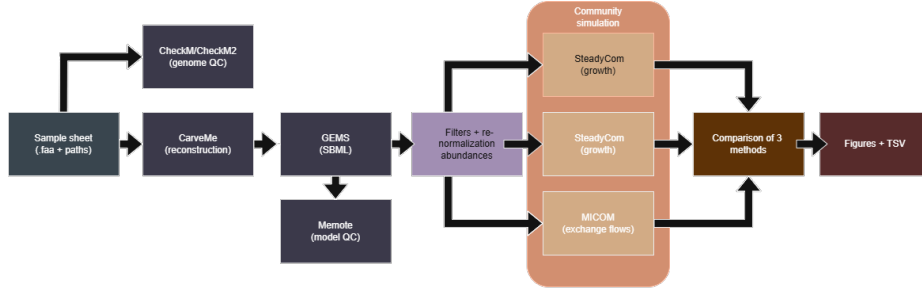


Fig. 1: General structure of the workflow. Starting from a sample sheet with the paths of the annotated genomes (.faa), CarveMe reconstructs the GEMs. Quality is assessed by CheckM2 (or CheckM) and the reconstructed models by Memote. After filtering and re-normalisation of the abundances, the approved models feed SMETANA, SteadyCom, and MICOM, whose results are integrated in the comparison stage.

3 Materials and methods

This section describes the architecture of the developed workflow, its stages, the datasets used, and the procedures for processing and validation. The code is organised according to the structure recommended by Snakemake, with a clear separation between rules (.smk files), scripts (Python), and software environments (Conda .yaml files), and is available, versioned, in the project repository.

3.1 Architecture and organisation of the workflow

The workflow is organised in a central Snakefile that defines the global variables and includes eight rule files, one per logical stage. The stages are: genome quality control with CheckM (or, in bypass mode, ingestion of a CheckM2 report); automatic reconstruction of GEMs with CarveMe; model validation with Memote; re-normalisation of relative abundances after quality filtering; the three community-simulation tools, SMETANA, SteadyCom, and MICOM; and a final comparison stage that integrates the outputs of the three tools. Each rule delegates the actual work to a dedicated Python script, keeping the execution logic separate from the definition of the graph. This modularity makes it possible to replace or adapt individual components without altering the rest of the flow. Figure 1 shows the overall structure.

A central design decision was the full parameterisation of the paths through the configuration file. The Snakefile reads from the configuration the input, results, and logs directories, with default values that reproduce the baseline behaviour. This makes it possible to run the same code over distinct datasets simply by providing a different configuration file, without any change to the code. Three configuration files were prepared: `config_reference.yaml` (reference genomes), `config_real.yaml` (22 real MAGs), and `config_real12.yaml` (the subset of

the 12 most abundant MAGs used for the detailed cross-feeding analysis; see Section 4.3).

3.2 Definition of the samples: sample sheet

The detection of samples follows a hierarchy of priorities implemented in the Snakefile: an explicit sample sheet (tabular format with `id` and `path` columns) takes precedence, followed by a list defined in the configuration and, lastly, automatic detection by file pattern. Using a versioned sample sheet rather than automatic detection confers explicit control over the set of analysed genomes and improves reproducibility. The following excerpt illustrates the reading of the sample sheet:

```
if SAMPLE_SHEET:
    with open(SAMPLE_SHEET) as f:
        reader = csv.DictReader(f, delimiter="\t")
        for row in reader:
            MAG_PATHS[row["id"].strip()] = row["path"].strip()
        MAGS = list(MAG_PATHS.keys())
```

3.3 Quality assessment: CheckM/CheckM2 and Memote

Genome quality assessment was designed to support two scenarios. In the first, applicable to genomes without prior assessment, quality is estimated within the pipeline with CheckM [8]. In the second, frequent in real scenarios, the assessment has already been carried out upstream (for example, with CheckM2 [22]) and is provided as a tabular report. For this case, the workflow integrates a bypass mode: instead of re-running CheckM, a dedicated rule extracts, for each genome, the corresponding line of the report and converts it to the internal format expected by the filtering step. This avoids unnecessary re-executions and, above all, preserves consistency with the quality criteria already applied by the team that generated the data. The selection of mode is automatic and depends on the presence of a report in the configuration: the bypass is activated whenever a `checkm_report` entry is present (that is, when `CHECKM_REPORT` is not `None`). The format conversion is tolerant to variations in column naming (for example, `Name` in CheckM2 and `Bin Id` in CheckM), ensuring robustness to different data provenances. After collecting the metrics, a filter retains only the genomes that satisfy the minimum completeness and maximum contamination thresholds defined in the configuration, in line with the MIMAG recommendations [3]. Complementarily, each reconstructed GEM is assessed with Memote [23], which produces a total score and per-section scores (stoichiometric consistency and annotation coverage). A second filter retains the models whose total score exceeds a configurable threshold, ensuring that only models of adequate quality proceed to the community analysis.

3.4 Model reconstruction: CarveMe

The GEMs are reconstructed with CarveMe [11] from protein files (.faa). As CarveMe targets prokaryotic genomes, the workflow is correspondingly oriented towards prokaryotic MAGs. The reconstruction script invokes the tool with the solver and culture medium defined in the configuration (CPLEX and M9 medium, respectively, in the real application), producing one model in SBML format per genome. The direct use of annotated protein files is particularly suitable for the application scenario, in which the MAGs had been previously annotated with Bakta [24], dispensing with an additional gene-prediction step.

3.5 Re-normalisation of abundances after filtering

In a real scenario, each MAG is associated with a relative abundance estimated by read mapping. Because the quality filters may remove some genomes, the effectively modelled community does not necessarily correspond to the initial set, and the original abundances (which sum to unity over the full set) would no longer be consistent. This step therefore requires, in addition to the MAGs, the table of per-MAG relative abundances as external input. To preserve coherence, a step recomputes the abundances after filtering: only the genomes simultaneously approved by the CheckM/CheckM2 and Memote criteria (intersection) are kept, and their abundances are re-normalised so that they again sum to unity. This ensures that the community tools receive a composition consistent with the genomes that actually enter the analysis, and traceably records which genomes were removed. The core logic is:

```
approved = checkm_pass & memote_pass
total_approved = sum(raw[s] for s in raw if s in approved)
renorm = {s: raw[s] / total_approved for s in approved}
```

3.6 Community-level simulation

The approved models are submitted to the three community tools. SMETANA is run in two modes: *global*, which returns the MIP and MRO indices of the community, and *detailed*, which enumerates the cross-feeding pairs and the shared metabolites. SteadyCom and MICOM are run through the MICOM library, receiving the table of re-normalised abundances. When this table is not provided (reference scenario), a uniform distribution is assumed. From MICOM, the exchange fluxes per organism are extracted and classified into produced and consumed metabolites. From SteadyCom, the growth rate and the steady-state abundance of each member are obtained.

3.7 Comparison between methods

The final stage integrates the results of the three tools into a per-organism table and a community-level synthesis. For each member, the growth rate (SteadyCom),

the counts of produced and consumed metabolites and the total exchange flux (MICOM), and the counts of donated and received metabolites (SMETANA) are gathered. The convergence between methods is quantified by Pearson correlation coefficients between analogous metrics, and the results are presented graphically.

3.8 Datasets and execution environment

The workflow was validated on a set of four reference genomes with well-characterised metabolism: *Bacillus subtilis* 168, *Escherichia coli* K-12, *Pseudomonas putida* NBRC 14164, and *Streptococcus pneumoniae* TIGR4. It was then applied to a real set of prokaryotic MAGs from an environmental sample (designated EST6). This sample originally yielded 74 prokaryotic bins; an upstream quality screening (completeness above 50% and contamination below 15%, consistent with the MIMAG medium-quality floor) retained 22 MAGs, which constitute the real dataset analysed here. These MAGs were annotated with Bakta [24], assessed with CheckM2 [22], and their relative abundances were estimated by read mapping with CoverM [26]. All executions took place in a Linux environment (WSL2/Ubuntu) with dependency management by Conda and the commercial CPLEX solver. The code is available in a public, Git-versioned repository.

4 Results

4.1 Validation on the reference set

The complete execution over the four reference genomes confirmed the chained operation of all stages, from the reconstruction of the GEMs to the community comparison. The reconstructed models obtained total Memote scores between 0.69 and 0.86, with the stoichiometric-consistency component ranging from 0.47 to 0.92; all models exceeded the defined quality threshold. At the community level, the global SMETANA indicated an MIP of 4 and an MRO of 0.65, and the detailed SMETANA identified 17 cross-feeding pairs involving 11 metabolites. In this controlled set, a strong convergence between independent methods was observed, with a Pearson correlation of 0.81 between the growth rate estimated by SteadyCom and the number of metabolites donated according to SMETANA, supporting the internal coherence of the pipeline.

4.2 Quality of the real MAGs

All 22 MAGs of the real dataset (selected as described in Section 3.8) were successfully reconstructed by CarveMe and approved by both the CheckM2 and Memote quality filters, thus all joining the modelled community. Table 1 summarises their completeness, contamination, and relative abundance. The genomes range from practically complete (several at 100% completeness) to medium quality. A completeness of 100% indicates that all expected single-copy

Table 1: Quality (CheckM2) and relative abundance of the 22 selected genomes, ordered by abundance. Relative abundances were estimated by read mapping with CoverM [26]. The dominant genome (**bin_020**) represents about 48% of the mapped community. For compactness, the 11 genomes individually below 2% abundance are summarised in the last row.

Bin	Completeness (%)	Contamination (%)	Abundance (%)
bin_020	93.3	1.22	47.96
bin_009	97.0	1.45	5.72
bin_055	99.2	5.13	4.65
bin_030	98.8	0.89	4.16
bin_001	100.0	3.54	3.99
bin_029	97.0	0.91	3.23
bin_040	100.0	3.21	2.88
bin_069	88.8	3.15	2.81
bin_056	96.4	3.02	2.73
bin_047	96.8	2.91	2.39
bin_023	88.9	9.52	1.99
(+ 11 bins, <2% each)	50.8–99.2	0.07–4.60	<2 each

marker genes were recovered, not that the genome is necessarily free of missing accessory genes. The community is strongly dominated by a single genome, **bin_020**, which represents about 48% of the mapped reads.

The abundance re-normalisation step behaved as expected: because none of the 22 genomes was removed by the filters, the re-normalised abundances continued to sum to unity and the dominance structure was preserved, with **bin_020** retaining its $\approx 48\%$ share.

4.3 Community simulation and cross-feeding

The detailed cross-feeding analysis with SMETANA has a computational cost that grows combinatorially with the number of organisms, and its execution over all 22 genomes proved infeasible within a reasonable time in the available environment. The detailed comparison was therefore restricted to the 12 most abundant genomes, which together represent approximately 85% of the mapped community, while the global indices (MIP/MRO) and the growth and flux modelling (SteadyCom and MICOM) were kept for the complete set of 22. This decision is discussed in Section 5.

The real community revealed a substantially denser interaction structure than the reference set. The global SMETANA indicated an MRO of approximately 0.72–0.73, stable between the 12- and 22-genome subsets, reflecting a high overlap of metabolic niches and therefore a strong potential for competition for resources. The detailed mode, over the 12 genomes, identified more than two hundred cross-feeding pairs, against only 17 in the reference set, an increase consistent with the greater diversity and dimension of the real community.

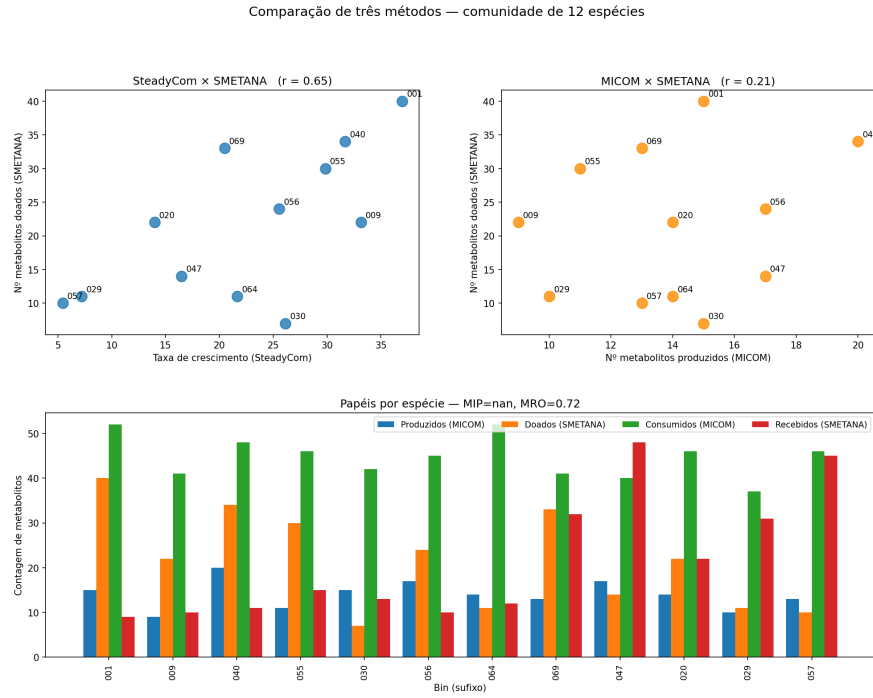


Fig 2: Comparison of the three methods in the real community (12 genomes). Top: scatter between the growth rate (SteadyCom) and the number of donated metabolites (SMETANA), and between the produced metabolites (MICOM) and the donated ones (SMETANA). Bottom: roles per organism according to each method (produced/consumed in MICOM; donated/received in SMETANA).

Unlike the reference set, the correlations between methods in the real community were weaker and, in some cases, divergent (Figure 2). The correlation between the growth rate (SteadyCom) and the donated metabolites (SMETANA) stood at around 0.5, while the correlation between the produced metabolites (MICOM) and the donated ones (SMETANA) was weak and unstable between executions. This divergence is interpretable in light of the conceptual differences between the tools: MICOM accounts for any secretion in a cooperative FBA solution (an inclusive criterion), whereas SMETANA classifies an organism as a donor only when its contribution is strictly necessary for another member (an exclusive criterion). In a simple community both criteria tend to coincide; in a dense community, with hundreds of possible interactions, the difference is amplified and the methods cease to agree.

4.4 Dissociation between abundance and metabolic centrality

A biologically notable result emerged from the per-organism analysis. The genome dominant in abundance (`bin_020`, $\sim 48\%$ of the community) is not the metabolically most active organism: it ranks among the lowest positions both in estimated growth rate and in total exchange flux. The correlation between relative abundance and estimated growth rate was in fact negative. In other words, the most abundant organisms do not necessarily correspond to the most active from a metabolic point of view, and comparatively rare genomes emerge as the main metabolic drivers of the community. This suggests that abundance (estimated by read mapping) and functional centrality (inferred from the metabolic models) are partially independent dimensions of the community structure.

5 Discussion

The results demonstrate that the workflow fulfils its objective of integrating, within a single and reproducible flow, the reconstruction of GEMs and the community simulation, being applicable both to reference genomes and to real MAGs. The validation on the reference set, with strong convergence between methods ($r = 0.81$), established a basis of confidence, and the application to the real data demonstrated the capacity of the pipeline to capture the ecological complexity of an environmental community.

The reduction of the detailed cross-feeding analysis to the 12 most abundant genomes deserves explicit justification. The cost of the detailed SMETANA grows with the square of the number of organisms (from 6 pairs for 4 genomes up to 231 for 22), rendering the complete execution impractical here. The abundance-based selection is reasonable ecologically, since the 12 retained genomes represent approximately 85% of the community and each excluded organism contributes less than 1.5%. This restriction applies only to the detailed analysis: the global cooperation/competition indices and the growth modelling were computed for all 22 genomes, so the objective of application to real MAGs was fully met.

The divergence observed between methods in the real community, far from being a weakness, is one of the analytical contributions of this work: it shows that the choice of simulation tool has a much greater impact in real, complex communities than in simple model systems. This reinforces the value of applying multiple methods over the same data to distinguish robust conclusions from methodological artefacts, and is consistent with structured tool evaluations reporting that different community-modelling tools can yield appreciably different predictions [13]. It should also be noted that SMETANA and MICOM incorporate stochastic components (multiple equivalent optima and sampling procedures), so the interaction metrics may vary slightly between executions; the values reported here correspond to a representative execution. The qualitative conclusions, namely the high metabolic overlap ($\text{MRO} \approx 0.72$), the density of cross-feeding, and the divergence between methods, remained consistent across executions, and a rigorous quantification of this variability through repeated executions is a natural extension of the work.

5.1 Limitations

The computational cost of the detailed SMETANA limits the exhaustive analysis of large-scale communities; more aggressive reduction or parallelisation strategies would be needed to encompass communities with many dozens of members. The absence of taxonomic assignment to the MAGs limits the direct biological interpretation of the results, and the integration of a classification step (for example, with GTDB-Tk [25]) would reinforce the ecological reading. Finally, automatic reconstruction with CarveMe, although fast, does not replace manual curation; the Memote scores, in particular in the stoichiometric-consistency component, indicate room for improvement in some models.

6 Conclusions

This work developed and applied a modular, reproducible, and documented workflow for the metabolic modelling of microbial communities from previously assembled genomes. The pipeline integrates, within a single Snakemake flow, genomic quality control, the automatic reconstruction of GEMs, their validation, and the simulation of community interactions with three complementary tools. The full parameterisation by configuration, the support for external quality reports, and the re-normalisation of abundances after filtering make the workflow directly applicable to real scenarios. The validation on reference genomes and the application to 22 environmental MAGs confirmed the robustness of the solution and produced biologically meaningful results, including a high metabolic overlap, a dense web of cross-feeding, and a dissociation between abundance and metabolic centrality. By bringing together multiple methods over the same data, the workflow helps to distinguish robust patterns from methodological artefacts, contributing to reduce the fragmentation that characterises the metabolic analysis of communities and providing a reusable basis for future studies.

References

1. Tyson, G.W., Chapman, J., Hugenholtz, P., Allen, E.E., Ram, R.J., Richardson, P.M., Solovyev, V.V., Rubin, E.M., Rokhsar, D.S., Banfield, J.F.: Community structure and metabolism through reconstruction of microbial genomes from the environment. *Nature* **428**(6978), 37–43 (2004). <https://doi.org/10.1038/nature02340>
2. Parks, D.H., Rinke, C., Chuvochina, M., Chaumeil, P.-A., Woodcroft, B.J., Evans, P.N., Hugenholtz, P., Tyson, G.W.: Recovery of nearly 8,000 metagenome-assembled genomes substantially expands the tree of life. *Nature Microbiology* **2**(11), 1533–1542 (2017). <https://doi.org/10.1038/s41564-017-0012-7>
3. Bowers, R.M., Kyrpides, N.C., Stepanauskas, R., Harmon-Smith, M., Doud, D., Reddy, T.B.K., Schulz, F., Jarett, J., Rivers, A.R., Elie-Fadrosh, E.A., et al.: Minimum information about a single amplified genome (MISAG) and a metagenome-assembled genome (MIMAG) of bacteria and archaea. *Nature Biotechnology* **35**(8), 725–731 (2017). <https://doi.org/10.1038/nbt.3893>
4. Zorrilla, F., Buric, F., Patil, K.R., Zelezniak, A.: metaGEM: reconstruction of genome scale metabolic models directly from metagenomes. *Nucleic Acids Research* **49**(21), e126 (2021). <https://doi.org/10.1093/nar/gkab815>
5. Köster, J., Rahmann, S.: Snakemake—a scalable bioinformatics workflow engine. *Bioinformatics* **28**(19), 2520–2522 (2012). <https://doi.org/10.1093/bioinformatics/bts480>
6. Mölder, F., Jablonski, K.P., Letcher, B., Hall, M.B., Tomkins-Tinch, C.H., Sochat, V., Forster, J., Lee, S., Twardziok, S.O., Kanitz, A., Wilm, A., Holtgrewe, M., Rahmann, S., Nahnsen, S., Köster, J.: Sustainable data analysis with Snakemake. *F1000Research* **10**, 33 (2021). <https://doi.org/10.12688/f1000research.29032.3>
7. Nayfach, S., Roux, S., Seshadri, R., Udway, D., Varghese, N., Schulz, F., Wu, D., Paez-Espino, D., Chen, I.-M.A., Huntemann, M., Palaniappan, K., Ladau, J., Mukherjee, S., Reddy, T.B.K., Nielsen, T., Kirton, E., Faria, J.P., Edirisinghe, J.N., Henry, C.S., Jungbluth, S.P., et al.: A genomic catalog of Earth’s microbiomes. *Nature Biotechnology* **39**(4), 499–509 (2021). <https://doi.org/10.1038/s41587-020-0718-6>
8. Parks, D.H., Imelfort, M., Skennerton, C.T., Hugenholtz, P., Tyson, G.W.: CheckM: assessing the quality of microbial genomes recovered from isolates, single cells, and metagenomes. *Genome Research* **25**(7), 1043–1055 (2015). <https://doi.org/10.1101/gr.186072.114>
9. Orth, J.D., Thiele, I., Palsson, B.Ø.: What is flux balance analysis? *Nature Biotechnology* **28**(3), 245–248 (2010). <https://doi.org/10.1038/nbt.1614>
10. Thiele, I., Palsson, B.Ø.: A protocol for generating a high-quality genome-scale metabolic reconstruction. *Nature Protocols* **5**(1), 93–121 (2010). <https://doi.org/10.1038/nprot.2009.203>
11. Machado, D., Andrejev, S., Tramontano, M., Patil, K.R.: Fast automated reconstruction of genome-scale metabolic models for microbial species and communities. *Nucleic Acids Research* **46**(15), 7542–7553 (2018). <https://doi.org/10.1093/nar/gky537>
12. King, Z.A., Lu, J., Dräger, A., Miller, P., Federowicz, S., Lerman, J.A., Ebrahim, A., Palsson, B.Ø., Lewis, N.E.: BiGG Models: A platform for integrating, standardizing and sharing genome-scale models. *Nucleic Acids Research* **44**(D1), D515–D522 (2016). <https://doi.org/10.1093/nar/gkv1049>

13. Scott Jr., W.T., Benito-Vaquerizo, S., Zimmermann, J., Bajić, D., Heinken, A., Suarez-Diez, M., Schaap, P.J.: A structured evaluation of genome-scale constraint-based modeling tools for microbial consortia. *PLoS Computational Biology* **19**(8), e1011363 (2023). <https://doi.org/10.1371/journal.pcbi.1011363>
14. Zelezniak, A., Andrejev, S., Ponomarova, O., Mende, D.R., Bork, P., Patil, K.R.: Metabolic dependencies drive species co-occurrence in diverse microbial communities. *Proceedings of the National Academy of Sciences of the United States of America* **112**(20), 6449–6454 (2015). <https://doi.org/10.1073/pnas.1421834112>
15. SMETANA documentation: Algorithms. <https://smetana.readthedocs.io/en/latest/algorithms.html>, last accessed 2026/04/05
16. SMETANA documentation: Usage. <https://smetana.readthedocs.io/en/latest/usage.html>, last accessed 2026/04/05
17. Chan, S.H.J., Simons, M.N., Maranas, C.D.: SteadyCom: Predicting microbial abundances while ensuring community stability. *PLoS Computational Biology* **13**(5), e1005539 (2017). <https://doi.org/10.1371/journal.pcbi.1005539>
18. Diener, C., Gibbons, S.M., Resendis-Antonio, O.: MICOM: Metagenome-scale modeling to infer metabolic interactions in the gut microbiota. *mSystems* **5**(1), e00606-19 (2020). <https://doi.org/10.1128/mSystems.00606-19>
19. Mesirov, J.P.: Accessible reproducible research. *Science* **327**(5964), 415–416 (2010). <https://doi.org/10.1126/science.1179653>
20. Sandve, G.K., Nekrutenko, A., Taylor, J., Hovig, E.: Ten simple rules for reproducible computational research. *PLoS Computational Biology* **9**(10), e1003285 (2013). <https://doi.org/10.1371/journal.pcbi.1003285>
21. Snakemake documentation. <https://snakemake.readthedocs.io/en/stable/>, last accessed 2026/04/05
22. Chklovski, A., Parks, D.H., Woodcroft, B.J., Tyson, G.W.: CheckM2: a rapid, scalable and accurate tool for assessing microbial genome quality using machine learning. *Nature Methods* **20**(8), 1203–1212 (2023). <https://doi.org/10.1038/s41592-023-01940-w>
23. Lieven, C., Beber, M.E., Olivier, B.G., et al.: MEMOTE for standardized genome-scale metabolic model testing. *Nature Biotechnology* **38**(3), 272–276 (2020). <https://doi.org/10.1038/s41587-020-0446-y>
24. Schwengers, O., Jelonek, L., Dieckmann, M.A., Beyvers, S., Blom, J., Goesmann, A.: Bakta: rapid and standardized annotation of bacterial genomes via alignment-free sequence identification. *Microbial Genomics* **7**(11), 000685 (2021). <https://doi.org/10.1099/mgen.0.000685>
25. Chaumeil, P.-A., Mussig, A.J., Hugenholtz, P., Parks, D.H.: GTDB-Tk: a toolkit to classify genomes with the Genome Taxonomy Database. *Bioinformatics* **36**(6), 1925–1927 (2020). <https://doi.org/10.1093/bioinformatics/btz848>
26. Woodcroft, B.J., et al.: CoverM: read coverage calculator for metagenomics. <https://github.com/wood/CoverM>, last accessed 2026/06/17